

FULL-STACK AI AGENT

Engineer

Complete Learning Roadmap — From Foundation to Production

CHAPTER 01

Full-Stack Foundation

01

Backend

Python
Core language
FastAPI
API framework
Node.js / TS
JS ecosystem
REST + WebSocket
Communication
PostgreSQL
Database
pgvector
Vector search
Redis
Caching / Queue
Kafka
Event streaming

02

Frontend

React / Next.js
UI framework
Tailwind CSS
Styling system
Shadcn / Radix
Component library
WebSocket / SSE
Real-time streaming
TypeScript
Type safety

03

DevOps

Docker
Containerization
Docker Compose
Multi-service
GitHub Actions
CI / CD
Linux Basics
Server mgmt
Nginx / Caddy
Reverse proxy

Foundation Layer — Prerequisite knowledge before entering AI Agent specialization

01

LLM Fundamentals

Prompt Engineering

System / few-shot / CoT

Tool Calling

Function call orchestration

Structured Output

JSON mode / grammar

Context Management

Window / summarize / sliding

Token Optimization

Cost & latency control

02

Agent Architecture

ReAct Loop

Thought → Action → Observe

Memory Systems

Short / long-term / RAG

Task Decomposition

Planner → Worker pattern

Tool Registry

Schema / validation / retry

Error Recovery

Auto-retry / fallback

03

Multi-Agent Systems

Comm Protocols

Message routing / broadcast

Orchestration

Sequential / Hierarchical

Debate & Voting

Multi-perspective consensus

Shared State

Blackboard / event bus

Supervisor Pattern

Manager → Specialist

04

RAG Full Pipeline

Chunking Strategy

Semantic vs fixed-size

Embedding + Vector Search

Dense / sparse / hybrid

Reranking

Two-stage precision

Query Rewriting

Multi-hop decomposition

Citation Grounding

Source attribution

05

Safety & Reliability

Sandbox Execution

Docker / Pyodide isolation

Guardrails

Output validation / filtering

Observability

Logging / tracing / cost

Rate Limiting

Token & request budgeting

Evaluation

Benchmarks / regression tests

Specialization Layer – Core competencies that define an AI Agent engineer